

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1188

Sherbino I 138 kV
April 02, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1188. The Interconnection Customer proposes a 274.2 MW solar plus storage facility interconnecting at the Sherbino I 138 kV substation in the West Virginia area. The request was received on April 13, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 314 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,189,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BERLIN CTR / FALLS CITY 138 KV	SHERBINO I / COAL 138 KV	232.7	187	100.0
HARRIMAN / FALLS CITY 138 KV	SHERBINO I / COAL 138 KV	227.1	194	100.0
HARRIMAN / SHERBINO I 138 KV	SHERBINO I / COAL 138 KV L	227.0	195	100.0
BERLIN CTR SUBSTATION / DEXTER TAP 138 KV	SHERBINO I / COAL 138 KV	165.7	261	100.0
S WINDSOR / DEXTER TAP 138 KV	SHERBINO I / COAL 138 KV	156.2	271	100.0
TILTON / S WINDSOR 138 KV	COAL / SHERBINO I 138 KV	154.4	273	100.0
OSO / TILTON 138 KV	SHERBINO I / COAL 138 KV L	127.0	204	57.4
CENTRAL JUNCOS 3202 SUBSTATION / RAWSON PACKAGING 138 KV L	MCCLAIN ENERGY FACILITY / COAL SUBSTATION 138 KV	123.0	289	73.3
COLTON / CENTRAL JUNCOS 3202 138 KV	MCCLAIN ENERGY FACILITY / COAL 138 KV	110.9	324	73.3
COAL / COLTON 138 KV	MCCLAIN ENERGY FACILITY / COAL 138 KV L	110.8	325	73.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BERLIN CTR / FALLS CITY 138 KV	Rebuild line with higher-capacity conductor	\$4,277,000	62% (\$2,633,000)
HARRIMAN / FALLS CITY 138 KV	Rebuild line with higher-capacity conductor	\$3,865,000	62% (\$2,379,000)
SHERBINO I / HARRIMAN SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$7,413,000	62% (\$4,563,000)
BERLIN CTR SUBSTATION / DEXTER TAP 138 KV	Rebuild line with higher-capacity conductor	\$6,520,000	62% (\$4,013,000)
S WINDSOR / DEXER TAP 138 KV	Rebuild line with higher-capacity conductor	\$10,597,000	62% (\$6,522,000)
SOUTH WINDSOR / TILTON 138 KV	Rebuild line with higher-capacity conductor	\$23,824,000	62% (\$14,663,000)
TILTN / OSO 138 KV Line	Rebuild line with higher-capacity conductor	\$4,505,000	62% (\$2,773,000)
RAWSON PACKAGING / CENTRAL JUNCOS 3202 138 KV	Rebuild line with higher-capacity conductor	\$3,709,000	60% (\$2,244,000)
COLTON / CENTRAL JUNCOS 3202 138 KV	Rebuild line with higher-capacity conductor	\$5,845,000	60% (\$3,537,000)
COLTON / COAL 138 KV	Rebuild line with higher-capacity conductor	\$6,383,000	60% (\$3,862,000)
	Total allocated to project		\$47,189,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.